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Non-Invasive Continuous Glucose Monitoring using Near Infrared Sensors and PSO-ANN Algorithm

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Abstract

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Keywords

, artificial neural network, calibration models, components, circuits, devices and systems, diabetes mellitus, near-infrared spectroscopy, non-invasive, particle swarm optimization, signal processing and analysis

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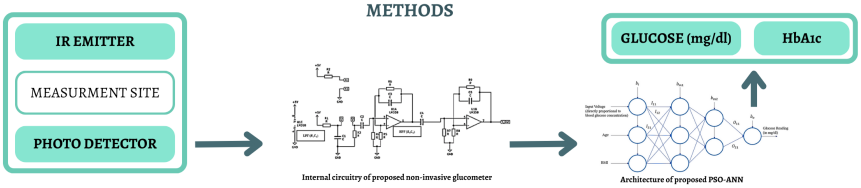
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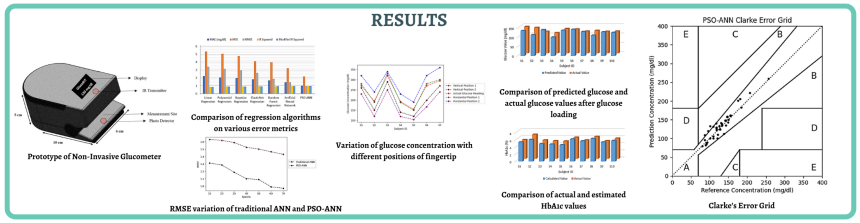
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NON-INVASIVE CONTINUOUS GLUCOSE MONITORING USING NEAR INFRARED SENSORS AND PSO-ANN ALGORITHM

METHODS



RESULTS



CONCLUSION

Our results show that the proposed PSO-ANN-based non-invasive glucometer performs competitively in comparison to commercial glucometers carrying additional benefits of HbA1c analysis and undergoing rigorous testing under OGTT. The paper also explores potential enhancements such as miniaturization of the prototype device for wearable applications, wireless connectivity, and integration with mobile devices for seamless data management and analysis.

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Abstract— Blood glucose is typically measured using invasive methods such as finger pricking, which although accurate are not suitable for frequent use as they cause extreme pain, and do not provide provisions for continuous glucose monitoring. Recent studies have proposed non-invasive glucometers that are based on scientific principles such as optical polarimetry, thermal emission, and electromagnetic approaches, but are expensive, highly sensitive to external noise and environmental variations, have low signal-to-noise ratio (SNR), and poor glucose selectivity. Although developments in Near-Infrared Spectroscopy (NIRS) have overcome these limitations to a certain extent, they do not produce reliable measurements due to large calibration errors that often result in incorrect glucose readings. In this paper, we propose a robust particle-swarm optimization-based artificial neural network for non-invasive continuous glucose monitoring using the principles of NIRS. We show that the PSO-ANN approach outperforms the traditional backpropagation algorithm used in ANN training and several other regression algorithms with the lowest error metrics: MAE- 1.01, MSE-2.16, RMSE-0.97, R^2-0.976 and modified R^2-0.973. The paper also provides insights into the circuit design, sensors used, hardware-software integration, and clinical validation alongside providing an overview of HbA1c computation. The accuracy and reliability of the proposed system are analyzed using the Clarke Error Grid (CEG) with 93.9% of the obtained readings falling within zone A and 100% of the readings falling in the clinically accepted range (zones A and B). The paper also explores potential enhancements such as miniaturization of the prototype device for wearable applications and wireless connectivity.

Index Terms— Diabetes mellitus, non-invasive, near-infrared spectroscopy, calibration models, artificial neural network, particle swarm optimization

I. INTRODUCTION

Diabetes mellitus (DM) is a metabolic disorder that occurs due to inefficient regulation of glucose levels in the body and is characterized by high glucose levels over a prolonged duration [1]. Frequent monitoring of blood glucose levels contributes to effective diabetic health management that helps identify fluctuations or anomalies in glucose concentration and further helps avoid health complications. The conventional finger pricking method is based on an electrochemical principle where small droplets of blood are placed on a test strip typically having glucose oxidase biosensors, and a chemical reaction on the strip results in a corresponding glucose reading [2]. Although conventional glucose measuring devices are accurate, they are not suitable for long-term use as they cause extreme pain, discomfort, and risk of infections. Alongside, the test strips are for single-time use and expensive, which limits their use to a maximum of 3 or 4 times a day. This on the other hand is not sufficient to keep track of an individual's diabetic health that needs continuous monitoring and analysis. Therefore, the development of non-invasive glucometers has garnered significant interest in recent years, aiming to provide a more convenient and user-friendly alternative for individuals with diabetes. Non-invasive methods can be broadly classified such as thermal, photo-acoustic, electro-chemical, electro-magnetic, optical, and biofluids-based approaches [3]. Among the various non-invasive techniques explored for glucose monitoring, Near Infrared Spectroscopy (NIRS) has emerged as a promising technology. NIRS utilizes the absorption and scattering properties of near-infrared light (wavelength in the range from 700 to 2500 nm) to obtain information about the molecular composition of biological tissues, including glucose concentrations. In this paper, a wavelength of 940 nm is selected that falls in the near-infrared region since glucose has a light absorption peak at this wavelength as well as the attenuation offered by other constituents of the blood like water, platelets, red blood cells, etc. is minimum at this wavelength [4]. This in turn helps achieve a desired depth of penetration and the actual glucose concentration can be predicted. Calibration of non-invasive glucometers is one of the most critical aspects that establishes a mapping between the absorbance of light and glucose concentration. This involves the application of Beer Lambert's law. In this paper, non-invasive glucose measurement is achieved using an infrared transmitter that emits waves in the NIR spectroscopy which is passed through the measurement site (typically fingertip or earlobe) and a photo-detector captures the amount of light absorbed by the tissues. The current flowing through the photo-detector is then converted to a voltage value using an operational amplifier which is further used in regression analysis. Recently, there have been several attempts to develop non-invasive glucose-measuring devices [6-9]. Although several studies in the literature report accuracies over 95% with clinical reliability of blood glucose readings, most of the proposed devices have not been tested under variations in glucose concentration. They have not been extensively tested under OGTT and glucose loading tests and do not convey information about parameters such as HbA1c which are important indicators of diabetic health. To bridge the gap between the current literature and the actual consumer requirement, we propose an end-to-end non-invasive glucometer based on NIRS that is calibrated to produce blood glucose readings based on a particle swarm optimized artificial neural network (PSO-ANN). We rigorously test the device under glucose loading tests and OGTT alongside providing provisions to compute HbA1c values that provide an estimate of the average blood glucose levels for the past three months. Further, we also discuss the design of our prototype which is made compact and portable for easy integration and deployment onto clinical systems and hand-held devices for remote glucose monitoring.

II. MATERIALS AND METHODS

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A. Beer Lambert's Law for Non-Invasive Glucose Measurement

Beer-Lambert Law states that the absorbance (A) of light passing through a sample is directly proportional to the concentration (c) of the absorbing substance and the path length (l) the light travels through the sample. Mathematically, it is expressed as shown in Eq. 1.

$$A = \varepsilon c l \tag{1}$$

where ε is the molar absorptivity, which represents the substance's ability to absorb light at a specific wavelength. From Eq. 1, it can be inferred that as glucose concentration increases, the absorbance increases proportionally. The increase in absorbance is further related to the attenuation of light given by Eq. 2.

$$I = I_0 e^{-\mu_{eff} L} \tag{2}$$

where I is the amount of light reflected, I_0 is the amount of light incident, L is the optical path length within the tissue and μ_{eff} is the effective coefficient of attenuation which is given by Eq. 3.

$$\mu_{eff} = 3\mu_a(\mu_a + \mu_s) \tag{3}$$

where μ_a is the coefficient of absorption and μ_s is the coefficient of scattering. With an increase in glucose concentration, the absorption of light increases which further increases its attenuation. By measuring the amount of attenuated light which is captured by a photodetector, the concentration of glucose molecules can be accurately measured which is the underlying principle of the proposed non-invasive glucometer. Light in the near-infrared region (0.7–5 µm) is considered in this work with a specific wavelength of 940 nm due to high glucose selectivity and the least attenuation offered by other blood constituents such as water and platelets at this wavelength. The amount of attenuation measured by the photodetector is further converted to an equivalent glucose value using an appropriate calibration algorithm and Eq. 4 depicts the general form of a glucose calibration algorithm.

$$\gamma = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \cdots + \beta_n x_n \tag{4}$$

where γ is the estimated glucose concentration, $\beta_0, \beta_1 \ldots \beta_n$ are the regression coefficients obtained during the calibration process and $x_1, x_2 \ldots\ldots x_n$ are the input features used in the calibration algorithm. For modeling a relation between the output of the photodetector and the blood glucose readings, we consider the input features as voltage (corresponding to glucose concentration), age, and BMI of the subject.

B. Three Stage Methodology

The overall methodology of continuous glucose monitoring is depicted in Fig. 1. The sensor unit consists of an Everlight IR-333A transmitter that emits infrared waves of wavelength 940 nm. This specific wavelength of 940 nm has maximum glucose sensitivity and the least attenuation offered by other constituents of blood and hence proves to be the most suitable wavelength in the NIR region for glucose measurement. The NIR waves emitted by the transmitter pass through the measurement site (fingertip) where light undergoes both absorption and scattering. The amount of light absorbed by the blood is measured by an Everlight phototransistor PT-333C that has a peak sensitivity at 940 nm. The photodetector measures this attenuation and a resultant current I flows through it which is further converted to a corresponding voltage value V using an operational amplifier. The signal processing unit consists of an operational amplifier, filter circuits, and an instrumentation amplifier. The output voltage V from the operational amplifier passes through a set of filter networks that consist of high-pass and low-pass filters where the filtered signal is then passed through an instrumentation amplifier. The output voltage V_{out} is fed to the data processing and analysis unit consisting of an ESP32 microcontroller integrated with an OLED unit for displaying corresponding glucose values. The microcontroller is programmed with the proposed PSO-ANN calibration algorithm. The algorithm is also designed to provide HbA1c values based on the measured blood glucose levels.

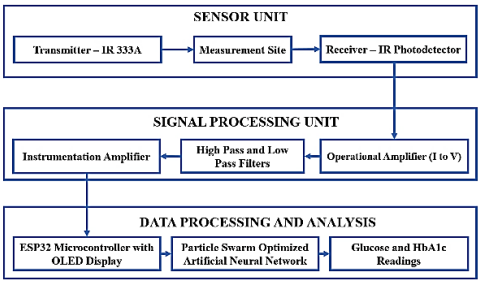


Fig. 1. Three-stage methodology of non-invasive glucometer

C. Circuit Design

The circuit employed in the proposed non-invasive glucometer is designed using KiCAD as shown in Fig. 2. It broadly consists of an IR light transmitter, photo-detector, filter networks, and amplification units. The current at the output of the photodetector is converted to a corresponding voltage value using an operational amplifier network with load resistance R_3 = 50k ohms at the anode. The cutoff frequencies for the low pass and high pass filters are chosen as 1.5 kHz and 2.2 kHz respectively while the voltage gain of the amplifier is chosen as 101 using Eq. 5, Eq. 6, and Eq.7.

The cutoff frequency for low pass filter (LPF) is calculated as:

$$\frac{1}{2\pi R_1 C_1} = \frac{1}{2*3.14*10^3*100*10^{-9}} = 1.5\ kHz \tag{5}$$

The cutoff frequency for high pass filter (HPF) is calculated as:

$$\frac{1}{2\pi R_2 C_2} = \frac{1}{2*3.14*68*10^3*10^{-6}} = 2.2\ kHz \tag{6}$$

The voltage gain of the amplification blocks in the circuit is given by Eq. 7.

$$A_0 = 1 + \frac{R_f}{R_m} = 1 + \frac{680*10^3}{68*10^3} = 101 \tag{7}$$

where A_0 is the output voltage gain.

The transmitter module consists of an IR 333A transmitter, a 5V supply, and a resistor $R_2 = 330$ ohms whereas the receiver module consists of a single low pass filter and two high pass filters coupled with two instrumentation amplifiers. The low pass filter consists of a resistor and a capacitor of values $R_1 = 1K$ ohms and $C_1 = 100nF$ respectively. The LPF is connected to the cathode of the photodiode. The load resistance $R_4 = 50k$ ohms is used to convert the output current from the photodiode into a corresponding voltage value. The high-pass filter consists of a resistor $R_2 = 68k$ ohms and a capacitor $C_2 = 1\mu F$, the output of which is connected to a LM358 instrumentation amplifier coupled with a RC feedback circuit with $R_6 = 680k$ ohms and $C_3 = 0.1\mu F$. The output of

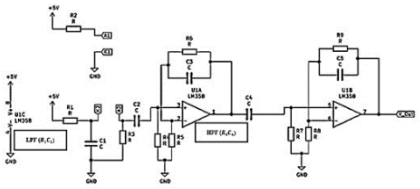


Fig. 2. Internal circuit design

LM358 is connected to a second HPF with resistance and capacitance values equal to $R_4 = 68k$ ohms and $C_4 = 1\mu F$ respectively. The output of the HPF is connected to a second LM358 instrumentation amplifier with an RC feedback circuit and the output voltage obtained at the final stage is denoted as V_{out}. This voltage is further fed to the V_{in} pin of the ESP32 microcontroller integrated with an OLED display module and programmed using the proposed PSO-ANN algorithm.

D. PSO-ANN Calibration Algorithm

The input features to the neural network are fed to the input layer which in our case is the output voltage V_{out}, age, and body mass index (BMI) of the subjects involved in this study. The weights assigned to each of these input features are updated by intermediate connections and weights in the hidden layer. We employ the sigmoid activation function in the intermediate layers that adjusts the weights of the intermediate layers and

help the weights propagate to the output layer. The weights of the intermediate layers are calculated using the particle swarm optimization (PSO) algorithm which is otherwise computed using back-propagation. PSO is a nature-inspired optimization that finds the most optimal solution for a given problem. It employs a S dimensional search space consisting of N randomly initialized particles in the population, each particle i in the swarm keeps track of the current position X_i, prior best position P_i and the current velocity V_i. Alongside the particles also keep track of the particle's best position P_g that is computed as a fitness value. Based on the value of the best position, each particle updates its current position and velocity given by Eq. 8 and Eq. 9 as depicted below.

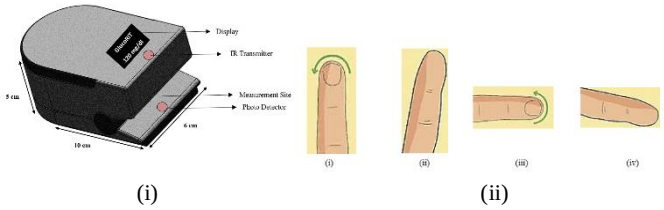
$$V_{new} = \alpha V_i + a_1(P_i - X_i)rand() + a_2(P_g - X_i)rand() \quad (8)$$

$$X_{new} = X_i + V_{new} \quad (9)$$

where V_{new} denotes the new particle velocity, a_1 and a_2 are acceleration constants (weighting coefficients), α is the inertial weight, $rand()$ is a random function in the range [0,1]. In this study, the major objective of the PSO algorithm is to minimize the error between the predicted glucose value and the actual glucose value given a set of input features.

E. Prototype Design of Non-Invasive Glucometer

We propose a compact, portable, and cost-effective design of a non-invasive glucometer depicted in Fig. 3 (i) that can be used at hospitals as well as by patients remotely. The glucometer has an arch-like structure with the OLED display mounted on the top surface whereas the transmitter and receiver LEDs are enclosed within the flaps (indicated by red circles). The prototype is designed using Autodesk Fusion 360 with the measurements indicated in the figure below. The measurement site indicates the location for the placement of the fingertip and Fig. 3 (ii) indicates different allowable positions of the fingertip for maximum accuracy of the device.



(i) (ii)

Fig. 3. (i) Prototype design of non-invasive glucometer (ii) Different positions for placement of fingertip (i) Vertical position 1 (ii) Vertical position 2 (iii) Horizontal position 1 (iv) Horizontal position 2

III. EXPERIMENTS AND RESULTS

A. Implementation Details of PSO-ANN Algorithm

In this study, we design an ANN with three neurons in the input layer, four neurons in the first hidden layer, five neurons in the second hidden layer, and one output neuron representing the estimated glucose value. We performed multiple experiments varying the number of neurons in the hidden layer and the 3-3-2-1 ANN architecture produced the most accurate results. This architecture involves a total of 17 weights to be optimized and in the PSO algorithm, we set the dimension of the search space to be 17. Table 1 depicts the performance comparison of different architectures of ANNs trained for 70 iterations each and the architecture with the least error and maximum accuracy is selected. The other major parameters in the PSO algorithm are the size of the population, acceleration constants, and inertial weight. We consider a population size of 25, acceleration constants a_1, a_2 and α as 2.76, 2.34, and 1.25 respectively as this combination of values yielded the lowest RMSE.

B. Performance Analysis of Regression Algorithms

We analyze the performance of different regression algorithms and evaluate them based on several metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), R^2 score and Modified R^2 score. Table 2 presents a performance comparison of several regression algorithms.

Table 1: Performance comparison of different ANN architectures

Architecture (with 2 hidden layers)	Validation Error (RMSE)	Accuracy (%)
3-3-3-1	2.451	65.1
3-4-3-1	2.109	69.6
3-4-4-1	1.897	75.4
3-5-4-1	1.658	78.0
3-5-5-1	1.432	84.8
3-4-5-1	0.970	98.2
3-2-2-1	0.964	98.5
3-3-2-1	**0.942**	**98.9**

Table 2: Performance comparison of different regression algorithms

Algorithm	**MAE (mg/dl)**	**MSE**	**RMSE**	**R^2**	**Modified R^2**
Linear Regression	2.23	5.32	3.39	0.841	0.840
Polynomial Regression	2.05	5.04	3.13	0.884	0.854
Bayesian Regression	1.98	4.76	2.95	0.890	0.890
ElasticNet Regression	1.82	4.09	2.62	0.927	0.911
Random Forest Regression	1.67	3.97	1.86	0.931	0.942
Artificial Neural Network	1.44	3.21	1.43	0.948	0.931
PSO-ANN	**1.01**	**2.16**	**0.97**	**0.976**	**0.973**

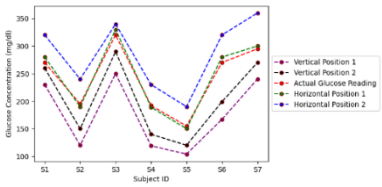
Table 3: Experimental results before glucose loading, after glucose loading, and random measurement

ID	Subject Age (years)	Voltage (mV) B/A/R	BMI	Before /After Glucose Loading		Random Glucose	
				Non-invasive	Invasive	Non-invasive	Invasive
S1	51	380/667/586	22.2	89.2/141.1	87.0/143.0	129.0	131.0
S2	45	545/620/574	21.9	112.0/136.5	110.0138.0	132.0	133.0
S3	20	368/419/541	18.9	118.2/118.3	120.0/117.0	118.0	117.0
S4	24	359/399/550	19.1	86.3/108.0	87.0/110.0	102.0	100.0
S5	38	568/570/581	23.4	133.8/131.0	132.0/132.0	130.0	129.0
S6	45	572/572/573	24.0	130.0/132.0	132.0/130.0	123.0	122.0
S7	31	507/556/582	22.7	116.9/119.1	118.0/118.0	102.0	103.0
S8	54	539/612/711	24.3	102.4/120.5	101.0/122.0	114.5	111.0
S9	23	371/360/345	18.3	109.3/120.9	110.0/117.0	107.0	105.0
S10	25	396/300/351	19.3	98.2/120.0	96.0/118.0	100.0	102.0

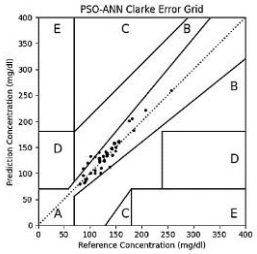
Through experiments, we find that the slope of the error curve for the PSO-optimized ANN is 28.7% lower than traditional ANNs. Hence, the proposed PSO-ANN algorithm proves to have higher reliability in non-invasive glucose measurement.

C. Clinical Trials - Glucose Loading and OGTT

In this study, we involved a total of 20 subjects belonging to the age group of 20 to 55 years who provided consent to participate in the glucose loading and random glucose tests. Out of the 20 subjects, 10 subjects participated in the glucose loading test and random glucose test, and the remaining 10 of them participated in HbA1c analysis. The entire experiment spanned for a total of 3 hours. We initially performed an oral glucose tolerance test (OGTT), where each subject's fasting glucose levels were measured. Each subject is asked to consume 5 g of glucose powder and their blood glucose levels are continuously monitored and measured at regular intervals of time. The glucose loading test was primarily performed to analyze and test the response of the device under varying levels of glucose concentration. This experiment was also done to measure the accuracy and sensitivity of the device to different concentrations of blood glucose. The results are summarized in Table 3.



(i)



(ii)

Fig. 4. (i) Variation of glucose concentration with different positions of fingertip (ii) Clarke's error grid

Fig. 4 (i) portrays the readings obtained for 7 subjects with different positions of the fingertip and the actual glucose reading for every subject measured using a commercial glucometer. It is seen that horizontal position 1 provides highly reliable and accurate glucose readings that are extremely close to the actual glucose readings.

Table 4: Experimental results for HbA1c comparison

ID	Subject Age (years)	Non-Invasive Glucometer (mg/dl)	Commercial Glucometer (mg/dl)	HbA1c (Actual) (mg/dl)	HbA1c (Estimated) (mg/dl)
S11	52	159.0	161.0	5.8	6.0
S12	49	139.0	138.0	4.8	4.5
S13	22	121.0	120.0	4.2	4.4
S14	25	123.0	124.0	4.0	4.3
S15	36	148.0	145.0	5.1	5.0
S16	45	131.0	130.0	5.2	5.4
S17	32	152.0	153.0	6.1	6.2
S18	53	216.0	211.0	6.9	7.1
S19	21	97.0	95.0	4.2	4.0
S20	20	100.0	102.0	4.7	4.8

D. HbA1c Computation and Analysis

Haemoglobin A1c (HbA1c) is a crucial parameter where the average blood glucose levels over the past two to three months are measured. It is an essential marker in diabetes management as it provides valuable information about a person's long-term glucose control. Using the estimated blood glucose levels non-invasively, we compute HbA1c as shown in Eq. 10 (a standard provided by the American Diabetes Association). The actual and estimated HbA1c values from the proposed device are listed in Table 4.

$$HbA1c = \frac{Glucose_{est}+46.7}{28.7} \quad (10)$$

The goal of Clarke's Error Grid is to ensure that most of the predicted glucose values fall into zones A and B, indicating accurate and clinically acceptable predictions. In this work, we obtain 100% of the estimated values to fall within Zones A and B with a total of 93.9% of the values falling in Zone A which is depicted in Fig. 4 (ii).

E. Comparison with Existing Implementations

In Table 5, we present a comparison of the proposed non-invasive glucometer's performance with several existing implementations. It can be inferred that our device shows robust performance on the metrics considered for comparison. The proposed glucometer has an added feature for HbA1c computation that most of the current devices lack. Our device has also been tested under several glucose loading tests and is calibrated optimally for robust real-time performance.

IV. CONCLUSION

In this paper, we present a NIRS-based non-invasive glucometer that is modeled using a PSO-ANN regression algorithm and has been extensively tested and analysed under varying conditions of glucose concentrations. The proposed hand-held compact device can be employed for continuous glucose monitoring in a pain-free and comfortable manner. With extensive experiments, we show that the proposed PSO-ANN algorithm achieves maximum accuracy, and minimum RMSE and converges to a global optimum much faster than the traditional BP-ANN. We provide an elaborate description of the circuit design, algorithmic specifications, and prototype design and provide a detailed explanation of different experiments conducted to test the reliability of the non-invasive glucometer. The proposed PSO-ANN algorithm outperforms the existing algorithms and we obtain 100% of the estimated glucose values to fall within the clinically acceptable regions. We also provide provisions for estimating HbA1c which most of the present approaches lack, and test the device under glucose loading and OGTT to provide robust performance under a range of conditions. In the future, the device will be integrated with miniaturized bio-wearable devices and will also be integrated with mobile devices.

Table 5: Performance comparison of the proposed device with existing approaches

Approach	RMSE	MAE	R^2	Clarke's Error Grid (Zone A+B)
[6]	6.89	-	-	100%
[9]	8.00	8.50	-	100%
[10]	-	-	0.910	100%
Ours	**0.97**	**1.01**	**0.976**	**100%**

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